Table 1: A feature comparison of EMBEDDING ATLAS against other popular embedding visualization tools. Legend: ● denotes a tool fully satisfies the feature, whereas ◐ denotes a tool partially satisfies the feature.

	Embedding Projector (2016)	**WizMap** (2023)	**Nomic Atlas** (2023)	**Data Map Plot** (2024)	**Latent Scope** (2024)	**EMBEDDING ATLAS** (2025)
F1. Fast interactive rendering of massive data	◐	●	●	●	●	●
F2. Multi-coordinated metadata views						●
F3. Automatic data clustering & labeling		◐	●		●	●
F4. Density contour rendering		●		●		●
F5. Real-time search & nearest neighbors	●	◐	◐	◐	◐	●
F6. Optional in-browser vector computation	●				◐	●
F7. Flexible data imports	●	◐	●	◐	●	●
F8. Extensibility		◐	◐		◐	●

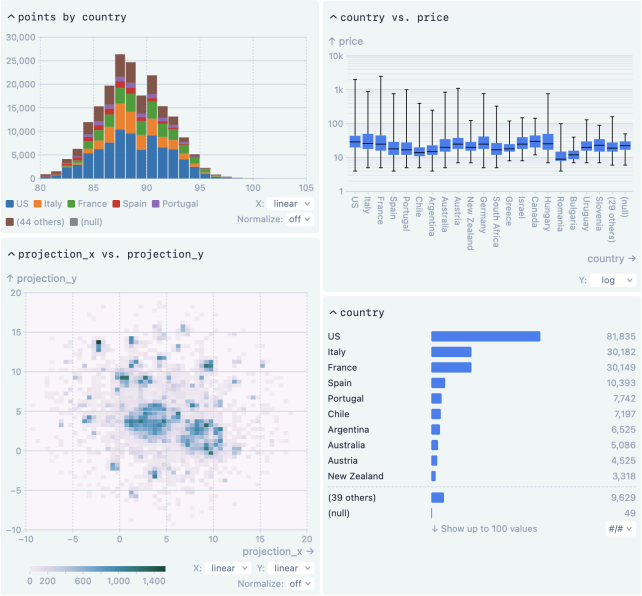

Figure 4: EMBEDDING ATLAS includes multiple distribution visualizations: bar charts, histograms, 2D heatmaps, and box plots. Basic charts can be added via a template-based chart builder. Advanced users can write arbitrary Mosaic specs to create complex charts.

4.2 Rendering Performance

We evaluate the rendering performance of the web-based embedding visualization by its frame rate. Figure 5 shows the results of an experiment with synthetic datasets of variable number of points and categories. The experiment is conducted with a 1600×1600 pixel embedding view (for 2x display scale), on an Apple M1 Pro with 32GB RAM, 10 core CPU, and 16 core GPU. EMBEDDING ATLAS can render up to 4 million points at 60fps, and maintain performance at 25fps for 10+ million points. With 1600×1600 pixels we did not notice a significant difference in FPS between points and density mode (Figure 3). However, since the density mode uses more involved pixel-space computation, its FPS drops as we increase the number of pixels (e.g., for 5 million points, FPS drops from 46 to 33 if we change to a 3840×2160 pixels view).

5 DISCUSSION

Flexible Data Processing with DuckDB and Mosaic EMBEDDING ATLAS supports both in-browser and server-side analysis via DuckDB and Mosaic [5], offering flexibility based on data size and user needs. For small datasets, run EMBEDDING ATLAS directly in the browser for a fast and installation-free experience. For larger datasets, EMBEDDING ATLAS connects to a server-side DuckDB instance to leverage more RAM and processing power.

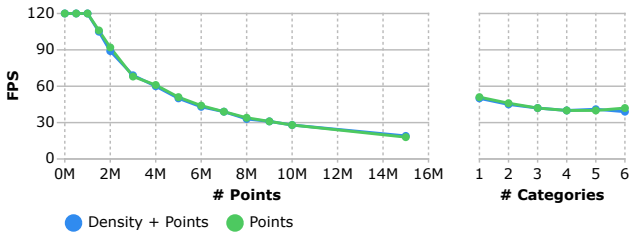

Figure 5: Left: Embedding view frames per second (FPS) over the number of points in the dataset. EMBEDDING ATLAS maintains above 60fps until 4 million points. Right: FPS over the number of categories for 5 million points.

Multiple Deployment Options For lighter use cases or quick exploration, EMBEDDING ATLAS has a web-based entrypoint that allows users to upload and visualize datasets directly in the browser with no installation required. For more advanced users, a command-line interface, Jupyter, and Streamlit widgets enable easy scripting and automation for complex workflows. Additionally, EMBEDDING ATLAS can be embedded as a component into into existing tools, dashboards, or research platforms.

6 LIMITATIONS AND FUTURE WORK

Embedding Comparison Comparing different datasets or dataset versions is often useful. While EMBEDDING ATLAS does not yet provide first-class comparison support, users can simulate it by adding auxiliary columns (e.g., dataset labels or version numbers) and using filtering or coloring. This approach has limitations (e.g., no side-by-side view), and first-class support would greatly enhance the comparison experience.

Towards a Dashboard-like Experience EMBEDDING ATLAS currently offers a single layout of the embedding view, charts, and the table. If there is no data specified for the embedding view, the tool stills shows the table and distribution charts; however, as we attempt to cover more use cases, this fixed layout becomes a limitation. In future versions, we would like to create a dashboard-like experience where users may freely design their own layout that best suit their analytical needs.

User Evaluation A user study is beyond the scope of this short paper. However, we plan to deploy the system to real-world data analysts, report its usage in practical settings, and collect feedback to guide future improvements. Since the tool is also open-sourced, we hope the community will guide future directions.

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